

TITLE: IMPLEMENTATION OF DDA (DIGITAL DIFFERENTIAL ANALYZER) ALGORITHM FOR LINE DRAWING

INTRODUCTION

The Digital Differential Analyzer (DDA) algorithm is a computer graphics technique used to draw straight lines on raster displays. It is a simple incremental scan-conversion method where pixel positions are calculated step-by-step between two given end points. The DDA algorithm works by calculating intermediate points along a line using the slope equation and plotting them, resulting in a smooth approximation of the actual line.

The main idea is to take unit steps in the direction of the dominant axis (either x-axis or y-axis), depending on whether the slope (m) of the line is less than or greater than 1.

- If $|m| \leq 1 \rightarrow x$ is incremented by 1 each step, and corresponding y is calculated
- If $|m| > 1 \rightarrow y$ is incremented by 1 each step, and corresponding x is calculated

This makes DDA efficient and easy to implement for drawing lines, polygons, and triangles.

ALGORITHM FOR DDA LINE DRAWING

Step 1: Start the program.

Step 2: Declare the variables: $x, y, x1, y1, x2, y2, k, dx, dy, s, xi, yi$ and also declare graphics driver $gdriver = DETECT$ and mode.

Step 3: Initialize the graphics mode using the proper path of the BGI folder in TurboC3.

Step 4: Input the two end-points of the line.

Step 5: Assign the left (starting) endpoint to $(x1, y1)$.

Step 6: Plot the first point $(x1, y1)$ using `putpixel()`.

Set $x = x1$ and $y = y1$.

Step 7: Calculate the difference values:

$$dx = x2 - x1$$

$$dy = y2 - y1$$

Step 8: Check if $|dx| > |dy|$

If true, then set $s = |dx|$

Else set $s = |dy|$

Step 9: Calculate the increment values:

$$xi = dx / s$$

$$y_i = dy / s$$

Step 10: Set $k = 0$

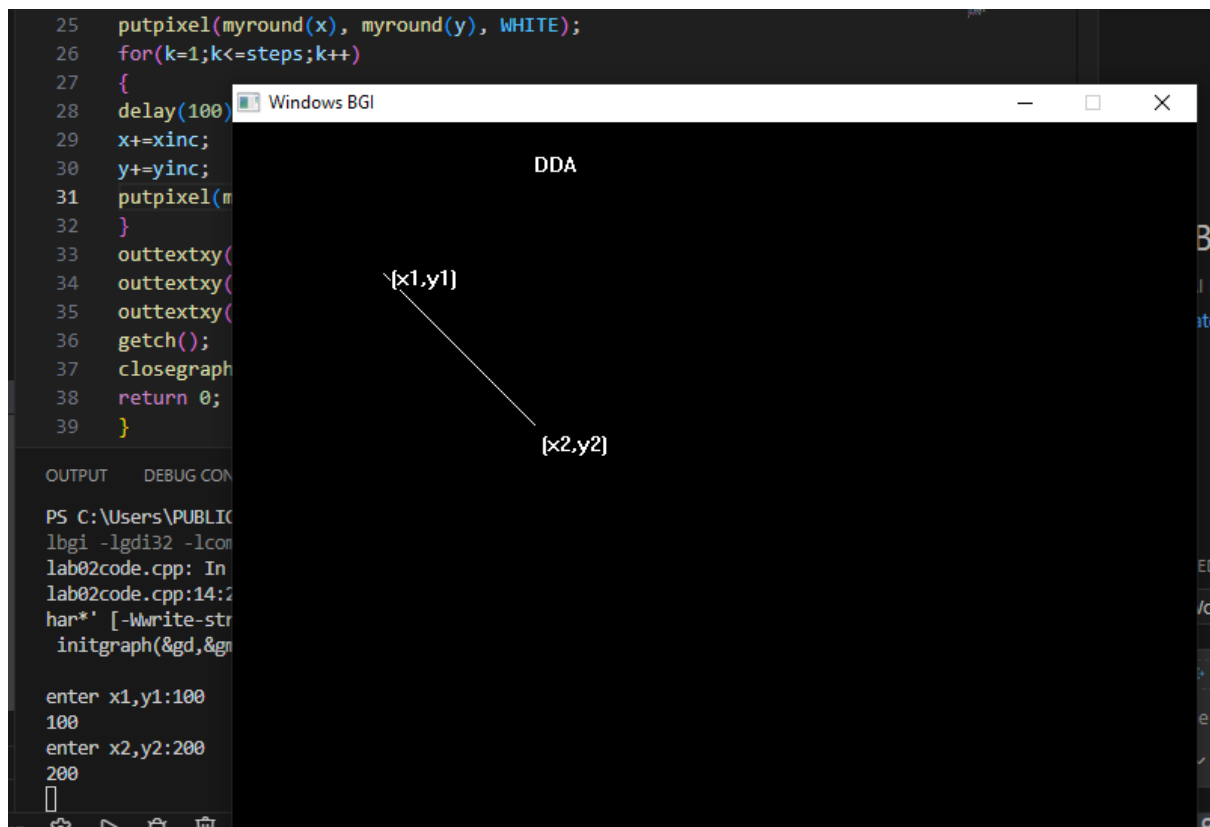
Step 11: Repeat while $k < s$

- $x = x + x_i$
- $y = y + y_i$
- Plot pixel at $(\text{round}(x), \text{round}(y))$ using `putpixel()`
- Increment k

Step 12: Close the graphics mode.

Step 13: Stop the program.

OUTPUT



CONCLUSION

The DDA algorithm is a simple and efficient method for drawing a line between two points in computer graphics. By incrementing the dominant axis and computing the corresponding value on the other axis, the algorithm produces a smooth rasterized line. Understanding DDA is essential because it forms the foundation for more advanced graphics algorithms used in modern rendering systems.